

History of AI

- 1943: early beginnings
 - McCulloch & Pitts: Boolean circuit model of brain
- 1950: Turing
 - Turing's "Computing Machinery and Intelligence"
- 1956: birth of AI
 - Dartmouth meeting: "Artificial Intelligence" name adopted
- 1950s: initial promise
 - Early AI programs, including
 - Samuel's checkers program
 - Newell & Simon's Logic Theorist
- 1955-65: "great enthusiasm"
 - Newell and Simon: GPS, general problem solver
 - Gelertner: Geometry Theorem Prover
 - McCarthy: invention of LISP

History of AI

- 1966—73: Reality dawns
 - Realization that many AI problems are intractable
 - Limitations of existing neural network methods identified
 - Neural network research almost disappears
- 1969—85: Adding domain knowledge
 - Development of knowledge-based systems
 - Success of rule-based expert systems,
 - E.g., DENDRAL, MYCIN
 - But were brittle and did not scale well in practice
- 1986-- Rise of machine learning
 - Neural networks return to popularity
 - Major advances in machine learning algorithms and applications
- 1990-- Role of uncertainty
 - Bayesian networks as a knowledge representation framework
- 1995-- AI as Science
 - Integration of learning, reasoning, knowledge representation
 - AI methods used in vision, language, data mining, etc

The Dartmouth Conference and the Name Artificial Intelligence

J. McCarthy, M. L. Minsky, N. Rochester, and C.E. Shannon. August 31, 1955. "We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."

Some of the applications are given below:

Business : Financial strategies, give advice

Engineering: check design, offer suggestions to create new product

Manufacturing: Assembly, inspection & maintenance

Mining: used when conditions are dangerous

Hospital : monitoring, diagnosing & prescribing

Education : In teaching

household : Advice on cooking, shopping etc.

farming : prune trees & selectively harvest mixed crops.

Present status

- Speech recognition : very accurate, uses hidden markov models(HMM)
- Planning : many very good planning systems used for scheduling of factory work, space missions etc.
- Dealing with uncertainty : Probabilistic reasoning used , Bayesian belief network (BBN)
- Decision theory and utility theory perfected
- Vast improvement in robotics, computer vision, machine learning and knowledge representation
- Lot of AI systems exist
- Deep learning has given a gigantic boost to AI

Recent AI

Heavy use of

- probability theory
- decision theory
- statistics
- logic (fuzzy, modal, temporal)

Intelligent Systems in Your Everyday Life

- **Post Office**
 - automatic address recognition and sorting of mail
- **Banks**
 - automatic check readers, signature verification systems
 - automated loan application classification
- **Customer Service**
 - automatic voice recognition
- **The Web**
 - Identifying your age, gender, location, from your Web surfing
 - Automated fraud detection
- **Digital Cameras**
 - Automated face detection and focusing
- **Computer Games**
 - Intelligent characters/agents

AI Applications: Machine Translation

- If you have a meeting of people from different nations for business like Japanese, Swedish etc.
- If you are shipping your software manuals to many countries
- You have to hire translators
- Cheaper and easier if the machine can do the translation
- Machine translation is hard for e.g.
 - English to Russian
 - “The spirit is willing but the flesh is weak” (English)
 - “the vodka is good but the meat is rotten” (Russian)
- However, there are many commercial systems that work well with algorithms using dictionaries, grammar models etc.
- Recently “black-box” machine learning techniques and deep learning techniques are doing well

Games

- 1950 Claude Shannon published a paper describing how a computer could play chess.
- 1952-1962 Art Samuel built the first checkers program
- 1957 Newell and Simon predicted that a computer will beat a human at chess within 10 years.
- 1967 MacHack was good enough to achieve a class-C rating in tournament chess.
- 1994 Chinook became the world checkers champion
- 1997 Deep Blue beat Kasparov
- 2007 Checkers is solved

Artificial Intelligence Vs. Conventional programming

• Dimension	Artificial Intelligence	Conventional Programming
• Processing	Primarily symbolic	Primarily algorithmic
• Nature of input	Can be incomplete	Must be complete
• Search	Heuristic (mostly)	Algorithmic
• Explanation	Provided	Usually not provided
• Major interest	Knowledge	Data, information
• Structure	Separation of control	Control integrated with
•	from knowledge	information (data)
• Nature of output	Can be incomplete	Must be correct
• Maintenance and	Easy because of	Usually difficult
• Update	modularity	
• Reasoning capacity	Limited but improving	None

Artificial Intelligence VS Conventional Computing

Artificial Intelligence Computing

1. AI software uses the techniques of search and pattern matching
2. Programmers design AI software to give the computer only the problem, not the steps necessary to solve it

Conventional Computing

Conventional computer software follow a logical series of steps to reach a conclusion

Computer programmers originally designed software that accomplished tasks by completing algorithms

Human Intelligence VS Artificial Intelligence

Pros

Human Intelligence

Intuition, Common sense,
Judgement, Creativity, Beliefs
etc

The ability to demonstrate their
intelligence by communicating
effectively

Plausible Reasoning and
Critical thinking

Artificial Intelligence

Ability to simulate human
behavior and cognitive
processes

Capture and preserve human
expertise

Fast Response. The ability to
comprehend large amounts of
data quickly.

Human Intelligence VS Artificial Intelligence

Cons

Human Intelligence

- Humans are fallible
- They have limited knowledge bases
- Information processing of serial nature proceed very slowly in the brain as compared to computers
- Humans are unable to retain large amounts of data in memory.

Artificial Intelligence

- No “common sense”
- Cannot readily deal with “mixed” knowledge
- May have high development costs
- Raise legal and ethical concerns

Human brain vs. Computers

- How complicated is our brain?
 - a neuron, or nerve cell, is the basic information processing unit
 - estimated to be on the order of 10^{12} neurons in a human brain
 - many more synapses (10^{14}) connecting these neurons
 - cycle time: 10^{-3} seconds (1 millisecond)
- How complex can we make computers?
 - 10^8 or more transistors per CPU
 - supercomputer: hundreds of CPUs, 10^{12} bits of RAM
 - cycle times: order of 10^{-9} seconds

Comparison of supercomputer and the human brain

	Supercomputer	Personal Computer	Human Brain
Computational units	10^4 CPUs, 10^{12} transistors	4 CPUs, 10^9 transistors	10^{11} neurons
Storage units	10^{14} bits RAM 10^{15} bits disk	10^{11} bits RAM 10^{13} bits disk	10^{11} neurons 10^{14} synapses
Cycle time	10^{-9} sec	10^{-9} sec	10^{-3} sec
Operations/sec	10^{15}	10^{10}	10^{17}
Memory updates/sec	10^{14}	10^{10}	10^{14}

Figure 1.3 A crude comparison of the raw computational resources available to the IBM BLUE GENE supercomputer, a typical personal computer of 2008, and the human brain. The brain's numbers are essentially fixed, whereas the supercomputer's numbers have been increasing by a factor of 10 every 5 years or so, allowing it to achieve rough parity with the brain. The personal computer lags behind on all metrics except cycle time.

From Russel and Norvig

Categories of AI System

- Systems that think like humans
- Systems that act like humans
- Systems that think rationally
- Systems that act rationally

Systems that think like humans

- We have to automate activities like decision making, problem solving, learning, etc.
- Most of the time it is a black box where we are not clear about our thought process.
- One has to know functioning of brain and its mechanism for possessing information.
- It is an area of cognitive science.
 - The stimuli are converted into mental representation.
 - Cognitive processes manipulate representation to build new representations that are used to generate actions.
- Neural network is a computing model for processing information similar to brain.

Systems that act like humans

- Computer should perform functions which are considered intelligent if performed by humans
- The overall behaviour of the system should be human like.
- It could be achieved by observation.

Systems that think rationally

- Use of mental faculties using computational models to perceive, reason and act
- Such systems rely on logic rather than human to measure correctness.
- For thinking rationally or logically, logic formulas and theories are used for synthesizing outcomes.
- For example,
 - given John is a human and all humans are mortal then one can conclude logically that John is mortal
- Not all intelligent behavior are mediated by logical deliberation.

Systems that act rationally

- Rational behavior means doing right thing
- Even if method is illogical, the observed behavior must be rational

Present state-of-the-art

- Robotic vehicles
 - A driverless robotic car STANLEY won the 2005 Darpa grand challenge
 - It was a Volkswagen Touareg car fitted with camera, radar and laser rangefinders and software to command the steering, breaking and acceleration
 - In 2006, CMU's boss won the urban challenge, driving safely, obeying traffic and avoiding pedestrians and other vehicles
- Speech Recognition
 - To book a flight, you can have the entire conversation guided by an automated speech recognition and dialog system
- Autonomous planning and scheduling
 - Use of onboard autonomous planning program to control the scheduling of operations of a spacecraft
 - Generating plans from high level goals and monitoring the execution of these plans – detecting, diagnosing and recovering from problems as they occur
 - Mission planning

Present state-of-the-art (contd.)

- Game playing
 - IBM's Deep Blue was first computer to defeat world champion Kasparov
 - Now human-computer matches are won convincingly by computers
- Spam classification and automatic deletion of spam
- Logistic planning
 - Automated logistic planning and scheduling for an large occurrence such as a war
 - Used in 1991 Gulf War
- Robotics
 - Roomba robotic vacuum cleaners for home made by iRobot Corporation
 - The same company used rugged bots in Iraq and Afghanistan to handle hazardous material, clear explosives and identify the location of snipers
- Machine Translation
 - Good machine translation programs exist today

Summary

- Interdisciplinary field
- Number of sub-areas
- Important field is knowledge based systems
- Uses non-monotonic reasoning
- Rigorous optimization and mathematics used for probabilistic reasoning, decision making
- A number of commercial packages available